

3D Scanning on Android Device System Requirements Document (SRD) Version 2 December 5th, 2025

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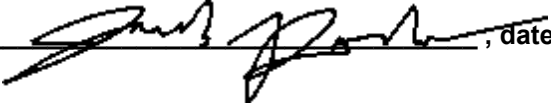
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1. Executive Summary

1.1 Project Overview

This project focuses on developing an Android application capable of generating 3D scans using the smartphone's integrated sensors, including the camera, accelerometer, and gyroscope. The application will capture real-time sensor data and apply Structure from Motion (SfM) algorithms to reconstruct a 3D point cloud of scanned objects. The primary purpose of this system is to assist prosthetic design by creating accessible and affordable scanning technology. The project is sponsored by ISO Technologies Inc., an organization that specializes in advanced materials for prosthetic manufacturing and insulation systems.

1.2 Purpose and Scope of this System Requirements Document

The purpose of this System Requirements Document (SRD) is to define all functional and non-functional requirements for the Android-based 3D scanning system. It will serve as a foundation for design, development, and validation.

In Scope:

- Image and motion data capture
- Point cloud generation
- Mesh reconstruction
- Real-time feedback
- Export to 3D formats (STL/OBJ)
- Measurement tools for prosthetic fitting

Out of Scope:

- GPU-intensive rendering
- Specialized hardware beyond phone sensors
- Multi-device synchronization.

1.3 Definitions, Acronyms, and Abbreviations

SfM – Structure from Motion

PCL – Point Cloud Library

STL/OBJ – Common 3D model export formats

IMU – Inertial Measurement Unit (accelerometer + gyroscope)

SDK – Software Development Kit

1.4 References

1. Project Charter: 3D Scanning on Android Device, ISO Technologies Inc, 2025
2. Android Camera2 API Documentation

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3. OpenCV Documentation
4. COLMAP Project Documentation
5. Point Cloud Library (PCL) Documentation
6. ISO Technologies Internal Technical Brief (2025)

2. Product/Service Description

2.1 Product Context

The 3D Scanning on Android Device application is a self-contained mobile solution that enables users to generate accurate 3D representations of physical objects. It uses the Android Camera2 API and sensor fusion from IMU data to produce a 3D point cloud through photogrammetry-based reconstruction. The final output can be exported to STL or OBJ formats for CAD/CAM integration and 3D printing. The app provides immediate visual feedback during scanning and operates fully offline.

2.2 Assumptions

- Users have access to a 2024 or newer Android device (12MP+ camera, 8GB RAM, Android 14+).
- Lighting during scan sessions is sufficient for consistent image quality.
- Users have basic familiarity with scanning workflow.
- Open-source vision libraries are compatible with target Android hardware.

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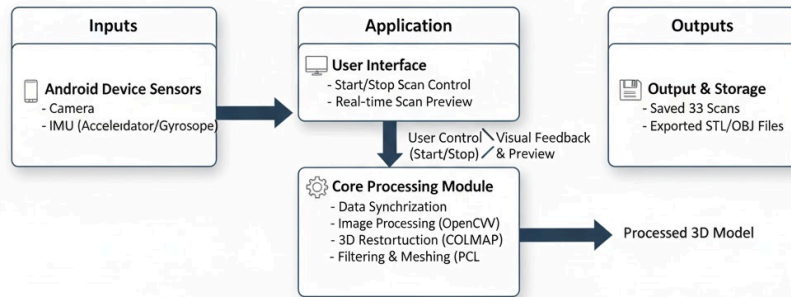


Figure 1: Application System Diagram

2.3 Constraints

- Accuracy target: $\pm 5\text{mm}$ for limb scans.
- Maximum app size: 100MB.
- Processing time: under 2 minutes per scan.
- Must function offline.
- No reliance on LiDAR or depth sensors.

2.4 Dependencies

- Android SDK and NDK
- Camera2 API
- OpenCV for image processing
- COLMAP for Structure from Motion
- PCL for point cloud filtering and meshing
- STL/OBJ file export utilities

3. Requirements

3.1 Functional Requirements

The following areas describe the functional and non-functional requirements of the system.

3.1.1 User Interface Requirements

- The application shall display a real-time preview of the scan area to the user.

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- The user interface shall provide feedback on scanning progress
- The user interface shall provide options to save a completed scan or export it to a 3D file format.

3.1.2 Performance

- Scan processing time shall be under 2 minutes for typical objects.
- The accuracy of the final 3D model for limb scans shall be within $\pm 5\text{mm}$.
- The application must function on mid-range Android devices (2024 or newer) as defined in the assumptions.

3.1.3 Capacity

- The application shall be capable of storing at least 10 completed scans on the local device storage.
- The final installed application size shall not exceed 100MB.

3.1.4 Systems Interfaces

- The application shall interface with the device's native camera API to capture high-resolution images.
- The application shall interface with the device's Inertial Measurement Unit (IMU) to capture synchronized accelerometer and gyroscope data.
- The system shall utilize a computer vision library for image processing tasks.
- The system shall utilize a Structure from Motion (SfM) library for 3D reconstruction.
- The system shall utilize a point cloud processing library for filtering and mesh generation.

3.2 System Requirements Matrix

Req #	Category	Requirement	Comments	Faculty Reviewed / Approved
1	Core Function	The system shall capture synchronized image and IMU data.	Uses the device's native camera and motion sensors.	
2	Core Function	The system shall generate a 3D point cloud from captured data.	This is the primary output of the SfM process.	
3	Performance	Scan processing shall be completed in under 2 minutes.	Applies to typical objects as defined in the use cases.	

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4	Performance	The final model shall have an accuracy of $\pm 5\text{mm}$ for limb scans.	Critical for the prosthetic design application.	
5	User Interface	The application shall provide real-time visual feedback during scanning.	Helps the user ensure complete data capture.	
6	Core Function	The system shall export the final model to STL and OBJ file formats.	For compatibility with CAD and 3D printing software.	
7	Availability	All core features shall function fully offline.	No reliance on cloud processing or internet connectivity.	
8	Capacity	The final installed application size shall not exceed 100MB.	Ensures accessibility for devices with limited storage.	

4. User Scenarios/Use Cases

The primary goal of this project is to address the high cost and complexity involved in manufacturing custom-fit prosthetics. The traditional method for creating a prosthetic socket relies on creating a plaster cast of a patient's limb, which is an involved process that can be uncomfortable for the patient and requires specialized, expensive equipment to digitize accurately. This financial and technical barrier can limit the availability of high-quality, custom-fit prosthetics in smaller or under-resourced clinics.

Use Case: Streamlining Prosthetic Socket Design

Consider a prosthetist in a local clinic who needs to design a custom socket for a patient. Instead of the traditional plaster casting method, the prosthetist uses the 3D scanning application on a standard smartphone. They open the app and, following the on-screen guides, move the device around the patient's residual limb. The real-time preview ensures a complete and accurate data capture is performed.

The entire scanning process is completed in minutes. The application then processes the captured images and sensor data to generate a precise 3D model. The prosthetist can inspect this model immediately, verify key

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measurements, and then export it as an STL file. This file is ready for use in CAD software or can be sent directly to a 3D printer for the fabrication of a test socket. In effect, the application replaces thousands of dollars of specialized equipment with a highly accessible software solution, making the process of creating custom prosthetics significantly faster, more efficient, and more affordable.

Use Case: Digital Archiving and Reproduction for Taxidermy

Beyond medical applications, the technology is also valuable for artists and preservationists, such as taxidermists. A finished taxidermy mount is a unique piece of art that can be fragile and susceptible to damage over time. Traditionally, repairing or reproducing such a piece is a difficult, manual process, and creating digital models requires expensive, specialized scanning equipment.

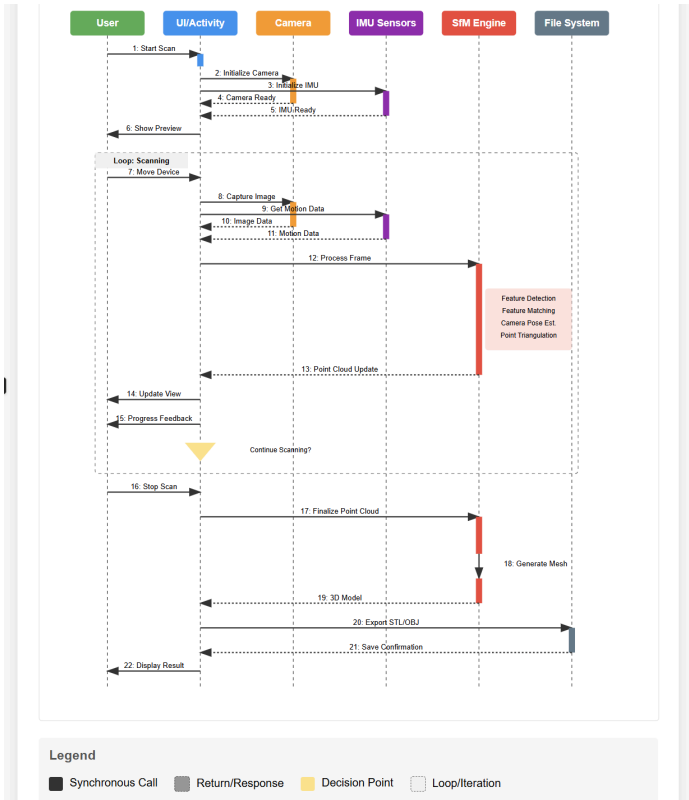
This application offers a practical solution. A taxidermist can use their smartphone to perform a detailed scan of a finished mount, like a deer head with a complex set of antlers. Following the on-screen guides, they can capture the geometry from all angles. The app processes the data and generates a high-fidelity 3D model. This digital file serves as a permanent archive of the piece. If the original mount is ever damaged, the 3D model can be used as a precise reference for restoration. Furthermore, the model can be sent to a 3D printer to create scaled-down replicas for clients or for educational purposes, opening up new creative and commercial possibilities without a major investment in hardware.

5. Analysis Models

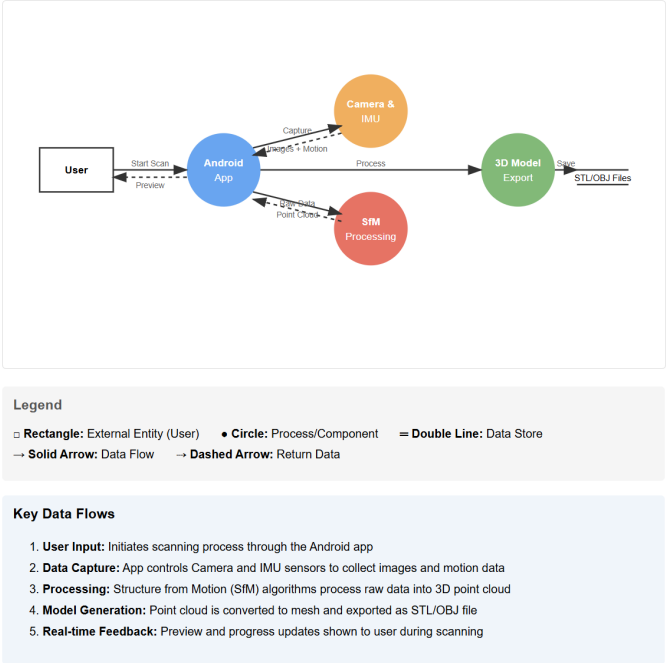
Sequence diagrams and data flow diagrams will model system interactions between sensors, processing modules, and UI components. The SWaP analysis shows negligible physical footprint since the system is entirely software-based. Power consumption is limited to standard smartphone use (<5W during active scanning).

5.1 System Sequence Diagram (Version 1 - Potential to Change)

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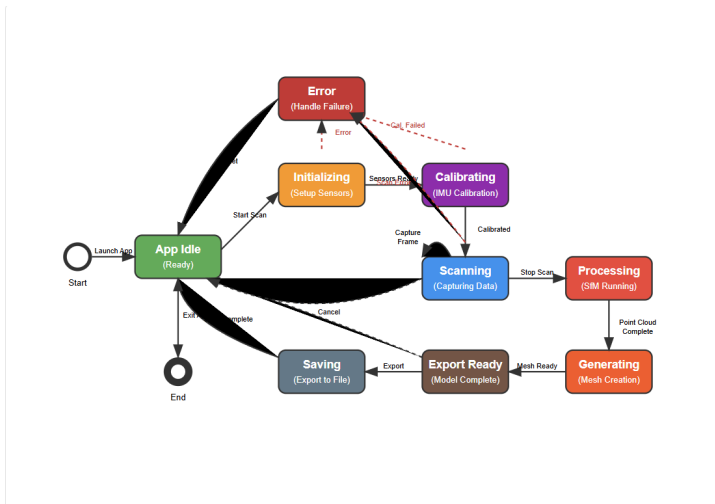


5.2 Data Flow Diagram (Version 1 - Potential to Change)



5.3 System State Diagram (Version 1 - Potential to Change)

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Current State	Event/Condition	Next State	Action
Start	App Launch	App Idle	Initialize UI
App Idle	User Starts Scan	Initializing	Setup Camera & IMU
Initializing	Sensors Ready	Calibrating	Start IMU Calibration
Calibrating	Calibration Complete	Scanning	Begin Data Capture
Scanning	Capture Frame	Scanning	Store Image & Motion Data
Scanning	User Stops Scan	Processing	Start SfM Processing
Processing	Point Cloud Complete	Generating	Create Mesh
Generating	Mesh Ready	Export Ready	Prepare Model
Export Ready	User Exports	Saving	Write STL/OBJ File
Saving	Save Complete	App Idle	Display Success
Any State	Error Occurs	Error	Log & Display Error
Error	Reset	App Idle	Clear Resources

5.4 SWaP

5.4.1 Size

- Size constraints based on data size; shall not exceed 300 MB to ensure accessibility

5.4.2 Weight

- Software based, weight is negligible

5.4.3 Power

- Should not significantly consume power from the phone when actively scanning

6. Requirements Test Matrix

1. Verify 3D model accuracy using calibration object comparison (Test).
2. Validate processing speed under 2 minutes (Test).
3. Confirm successful export to STL/OBJ (Demonstration).
4. Inspect compatibility with Android 14+ devices (Inspection).

7. Project Risks

-Hardware Limitations: Low-end devices may struggle with processing large scans.
Mitigation: Optimize algorithms and use adaptive resolution settings.

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-Lighting Variability: Poor or inconsistent lighting can cause scan artifacts.

Mitigation: Apply preprocessing filters and guide users on proper lighting.

-Sensor Calibration: IMU or camera drift may lead to inaccurate models.

Mitigation: Implement automated and periodic recalibration.

-User Error: Incorrect scanning movement or angles may distort results.

Mitigation: Provide clear in-app tutorials and real-time feedback.

8. Standards

-STL/OBJ Export Formats:

Standardized 3D file formats for compatibility with CAD software, 3D printers, and prosthetic modeling tools.

[IEEE Standards Association+2IEEE 802+2](#)

-OpenCV and PCL Libraries:

Follow BSD open-source licensing and documentation standards for image capture, point-cloud generation, and 3D reconstruction.

-USB 3.0 / UART Communication Protocols:

Ensure reliable, high-speed data transfer between sensors, microcontrollers, and processing units.

<https://ieeexplore.ieee.org/book/9218891>

-ISO/ASTM 52900 Additive Manufacturing Standards:

Maintain consistency with international terminology and data formatting for 3D scanning and printing applications.

[ISO+2Antpedia+2](#)

9. Engineering Ethical Responsibility

9.1 Public Health

This project enhances public health by increasing access to custom prosthetic solutions. Using smartphone-based 3D scanning, clinics in underserved areas can generate accurate limb models quickly and affordably, enabling faster production of personalized prosthetics.

9.2 Safety

The system poses no direct physical risks. Ethical safety considerations center on the accuracy and precision of scans to ensure comfortable and properly fitted prosthetics. Calibration and validation procedures are incorporated to minimize measurement errors.

9.3 Global, Social, & Cultural Factors

By removing the need for expensive scanning hardware, this technology promotes inclusivity across different socioeconomic and cultural settings. It supports global access to assistive devices, allowing clinics and small workshops worldwide to implement low-cost digital fabrication.

9.4 Economic Factors

The mobile-based approach minimizes hardware waste and power consumption, extending the lifespan of existing devices. It reduces barriers for small clinics and startups, encouraging local innovation and sustainable growth in healthcare technology.

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9.5 Environmental Factors

The software-only approach minimizes hardware waste and energy consumption. By extending phone utility instead of creating new devices, it indirectly supports environmental sustainability.

10. Change Management Process

All proposed modifications to the project will be tracked using a shared version-controlled repository, such as GitHub. Team members will create and review change requests that describe the reason and impact of each update. Each revision will be documented with timestamps, author names, and a short summary to ensure accountability. Once verified, approved changes will be merged into the main branch after faculty advisor confirmation. This process ensures transparency, prevents version conflicts, and maintains a clear history of all project revisions.